North Ridge Highway Intersection Upgrade

Owner design brief

1. Commission and scope

North Ridge Industrial Lands Partnership commissions Cairnfield Civil Studio Ltd. for project NRH-26104, North Ridge Highway Intersection Upgrade. The rural two-lane provincial corridor serves an expanding industrial area in central Alberta. Scope includes turning-lane geometry, profile review, access, pavement and construction traffic accommodation.

The project uses a 100 km/h preliminary design speed and a 90 km/h existing posted speed. The owner adopts 185 m as the initial stopping-sight-distance screening value. Final highway classification, grade effects and any design exception require review with the provincial road authority.

2. Performance and deliverables

Opening year is 2027. Review the combined development and background traffic at 10- and 20-year horizons. The proposed work must retain access and maintain traffic during construction, subject to an accepted traffic accommodation plan.

Deliverables: survey/geotechnical summary, G101/G102/G201 geometric plan and profile, traffic impact assessment, pavement memorandum and traffic accommodation concept. No design exception or construction authorization is included in this issue.

Project	Issue information
NRH-26104	2026-08-14
North Ridge Industrial Lands Partnership	Issued for review
Prepared / checked	JL / AK

North Ridge Highway Intersection Upgrade

Owner design brief

3. Design basis and coordination

Use the current applicable Alberta Highway Geometric Design Guide chapters, Traffic Impact Assessment Guidelines, Pavement Design Manual together with Design Bulletin 77 amendments, and the 2025 Traffic Accommodation in Work Zones manual.

Use a reconciled base-year traffic dataset across traffic and pavement disciplines. Differences in forecasting method must be justified; traffic volumes, truck definitions, axle loading and design-lane allocation must be traceable before pavement selection.